

Asst.Prof.Dr.Wandee Udomuksorn

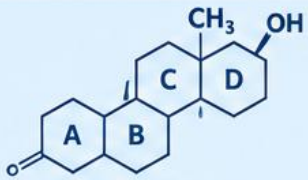
Program of Pharmacology

Faculty of Science, PSU

Objectives

- Definition of synthetic steroid
- Classification of synthetic steroids using in clinical
- Pharmacology of synthetic steroids
  - Glucocoticoids
  - Mineralocorticoids
  - Androgenic steroids
  - Anabolic steroids
  - Estrogenic steroids
  - Progestin (Progestogens)
- Pharmacology of steroid hormone receptor antagonists
- Pharmacology of steroid synthesis inhibitors

Definition and classification of synthetic steroids



Basic steroid nucleus (19-carbon skeleton)

Classification of Synthetic Steroids

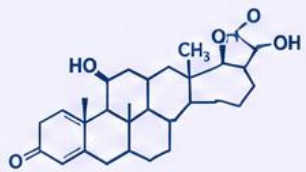
Synthetic steroids are classified based on their chemical structure and pharmacologic properties, which determine their therapeutic uses and side effects.

1. Glucocorticoids

Structural modification at C9, C11, C17 (and sometimes C16) gives strong glucocorticoid activity with minimal mineralocorticoid effect.

Examples

- Hydrocortisone
- Prednisone
- Dexamethasone
- Methylprednisolone



Dexamethasone

Key features / uses

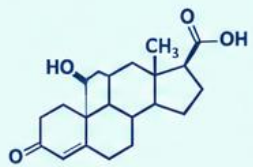
- Anti-inflammatory
- Immunosuppressive
- Used in autoimmune diseases, allergies, transplant rejection, etc.

2. Mineralocorticoids

Have strong activity on the mineralocorticoid receptor (aldosterone-like effect) with little glucocorticoid activity.

Examples

- Fludrocortisone
- Deoxycorticosterone (DOC) (less commonly used)



Fludrocortisone

Key features / uses

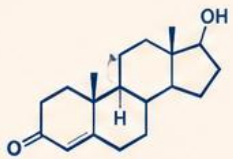
- Sodium and water retention
- Potassium excretion
- Used in adrenal insufficiency (e.g., Addison's disease)

3. Androgenic Steroids

Modified at C17 (usually 17α-alkylation) to increase oral activity and reduce metabolism.

Examples

- Testosterone
- Methyltestosterone
- Oxandrolone
- Stanozolol



Testosterone

Key features / uses

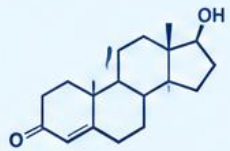
- Promote male secondary sexual characteristics
- Increase muscle mass and strength
- Used in hypogonadism, delayed puberty, wasting diseases

4. Anabolic Steroids

Preferentially enhance protein synthesis and muscle growth with less androgenic effects (especially with structural modifications).

Examples

- Nandrolone
- Stanozolol
- Oxandrolone
- Boldenone



Nandrolone

Key features / uses

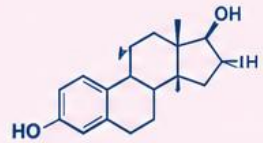
- Increase protein synthesis
- Build lean body mass
- Used in muscle wasting, osteoporosis, and performance enhancement (abuse)

5. Estrogenic Steroids

Modified at C3 and C17 to increase estrogenic activity (e.g., aromatic A ring or 17β-hydroxyl group).

Examples

- Ethinyl estradiol
- Estradiol
- Diethylstilbestrol (DES)
- Premarin (conjugated estrogens)



Ethinyl estradiol

Key features / uses

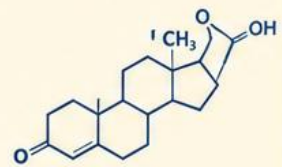
- Regulate menstrual cycle
- Contraception
- Hormone replacement therapy
- Certain cancers (e.g., breast cancer – DES)

6. Progestin (Progestogens)

Modified at C17 to have progestational activity (without significant estrogenic or androgenic effects).

Examples

- Medroxyprogesterone acetate
- Norethisterone
- Levonorgestrel
- Drospirenone



Medroxyprogesterone acetate

Key features / uses

- Support pregnancy
- Contraception
- Treat endometriosis, abnormal uterine bleeding

In summary: Synthetic steroids are grouped into glucocorticoids, mineralocorticoids, androgenic steroids, anabolic steroids, estrogenic steroids, and progestins, based on their structure and primary pharmacologic effects.

Pharmacokinetics

1. Absorption

- Most synthetic steroids are **lipid-soluble** and readily cross biological membranes.
- **Oral** administration: absorbed from the gastrointestinal tract, but the extent varies between drugs.
- **Intramuscular** injection: depot formulations can provide slow, prolonged absorption.
- **Topical/inhaled** routes: can produce mainly local effects, although some systemic absorption occurs.
- **Chemical modifications** can increase oral bioavailability and prolong activity.

Pharmacokinetics

2. Distribution

- After entering the bloodstream, steroids are distributed widely because of their **high lipid solubility**.
- They can cross cell membranes and enter target tissues.
- Many steroids bind to plasma proteins, mainly **albumin** and, for some corticosteroids, **corticosteroid-binding globulin**.
- Protein binding influences the **free (active) concentration** and **duration of action**.

Pharmacokinetics

3. Metabolism

- Metabolism occurs predominantly in the liver.
- Steroids undergo reactions such as oxidation, reduction, and conjugation.
- Synthetic modifications can make steroids more resistant to metabolism and therefore increase their duration of action.
- Some orally active anabolic steroids, particularly 17α-alkylated compounds, have increased oral activity but can also increase the risk of hepatic toxicity.

Pharmacokinetics

4. Excretion

- Steroids are converted into more water-soluble metabolites.
- These metabolites are eliminated mainly through the kidneys in urine.
- Some metabolites may also be excreted through bile and feces.

Pharmacokinetics

5. Half-life and duration of action

- The half-life varies greatly among synthetic steroids and depends on:
  - Chemical structure
  - Route of administration
  - Protein binding
  - Hepatic metabolism
  - Formulation, particularly **esterified/depot** preparations

Examples:

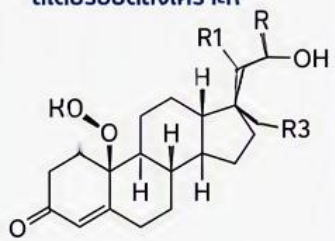
- **Hydrocortisone:** relatively short acting
- **Prednisolone:** intermediate acting
- **Dexamethasone:** long acting
- **Long-acting injectable steroid esters** may remain active for substantially longer because the formulation **slowly releases** the steroid.

Key point

- **Synthetic modifications of the steroid structure are used to change PK properties**
  - **Oral bioavailability, metabolic stability, half-life, tissue distribution, and duration of action**.
- These changes help determine which steroid is appropriate for a particular therapeutic purpose.

กลไกการออกฤทธิ์ของ SYNTHETIC STEROIDS

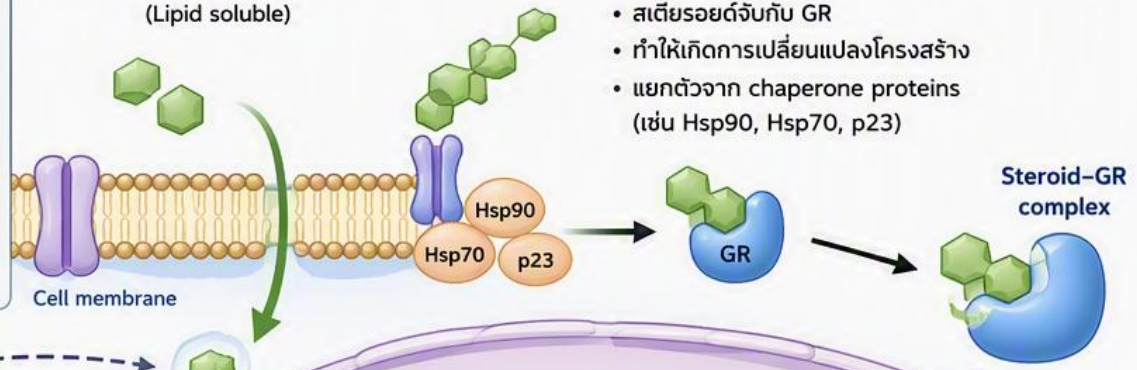
1 โครงสร้าง
สเตียรอยด์สังเคราะห์



Corticosteroids
(หมู่ R ต่างกันตามชนิดและความแรง)

2 ผ่านเข้าสู่เซลล์
- แพร่ผ่านเยื่อหุ้มเซลล์ (Lipid soluble)

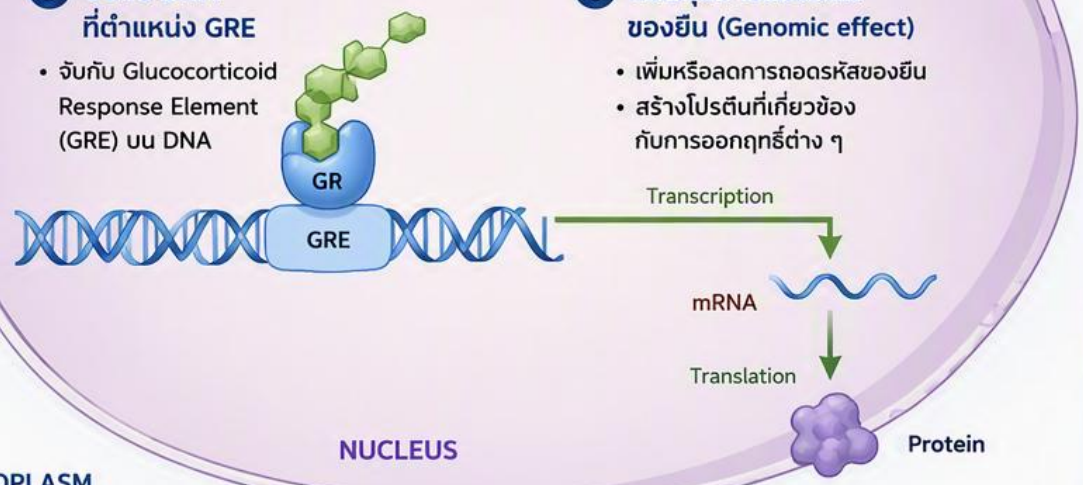
3 จับกับตัวรับภายในเซลล์ (Glucocorticoid Receptor, GR)
- สเตียรอยด์จับกับ GR
- ทำให้เกิดการเปลี่ยนแปลงโครงสร้าง
- แยกตัวจาก chaperone proteins (เช่น Hsp90, Hsp70, p23)



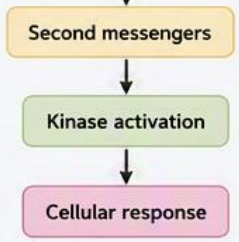
4 เคลื่อนเข้าสู่นิวเคลียส
- Steroid–GR complex เคลื่อนเข้าสู่นิวเคลียส

5 จับกับ DNA ที่ตำแหน่ง GRE
- จับกับ Glucocorticoid Response Element (GRE) บน DNA

6 ควบคุมการถอดรหัสของยีน (Genomic effect)
- เพิ่มหรือลดการถอดรหัสของยีน
- สร้างโปรตีนที่เกี่ยวข้องกับการออกฤทธิ์ต่าง ๆ



7 ฤทธิ์นอกเหนือจากการควบคุมยีน (Non-genomic effect)
- เกิดขึ้นรวดเร็วภายในไม่กี่วินาทีถึงนาที
- ผ่านตัวรับที่ผิวเซลล์หรือโมเลกุลอื่น เช่น membrane GR, GPR, ion channel
- กระตุ้นสัญญาณภายในเซลล์ เช่น cAMP, Ca^{2+}, MAPK, PI3K/Akt



ผลการออกฤทธิ์ (Pharmacologic effects)

ต้านการอักเสบ (Anti-inflammatory)
- ลดการสร้าง cytokines, COX-2, iNOS, adhesion molecules

กดภูมิคุ้มกัน (Immunosuppressive)
- ลดการทำงานของ T cells, B cells, macrophages, dendritic cells

ลดอาการแพ้ (Anti-allergic)
- ลดการหลั่งสารก่อภูมิแพ้และการซึมผ่านของหลอดเลือด

ผลต่อเมตาบอลิซึม (Metabolic effects)
- เพิ่มการสร้างกลูโคส, สลายโปรตีน, สะสมไขมัน

ผลต่อกระดูก (Bone effects)
- ลดการสร้างกระดูก, เพิ่มการสลายกระดูก

ผลต่อดวงตา (Ocular effects)
- เพิ่มความดันในลูกตา, ต้อกระจก

กดการทำงานของต่อมหมวกไต (HPA axis suppression)
- ลดการหลั่ง ACTH และ cortisol

ปัจจัยที่มีผลต่อความแรงของสเตียรอยด์
- ชนิดของหมู่ R ที่ตำแหน่งต่าง ๆ ในโครงสร้าง
- ความสามารถในการจับกับ GR
- ความสามารถในการเข้าสู่เซลล์
- อัตราการสลายและการกำจัด

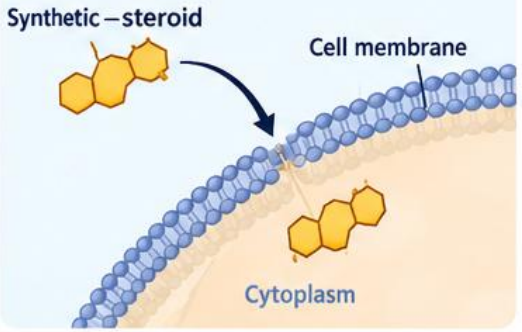
สรุป: สเตียรอยด์สังเคราะห์ผ่านเข้าสู่เซลล์ → จับกับตัวรับ GR → เข้าสู่นิวเคลียส → ควบคุมการแสดงออกของยีน → เกิดผลทางชีวภาพต่าง ๆ นอกจากนี้ยังสามารถออกฤทธิ์แบบไม่ผ่านการควบคุมยีนได้อย่างรวดเร็วผ่านสัญญาณภายในเซลล์

Mechanism of Synthetic Steroids

Synthetic steroids (anabolic-androgenic steroids, glucocorticoids, etc.) are designed to mimic natural steroid hormones and act through intracellular receptors to produce biological effects.

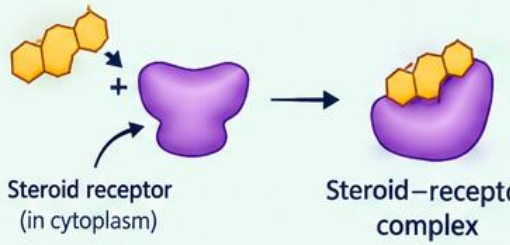
1 Enter the cell

Synthetic steroids are lipophilic, so they diffuse through the cell membrane.



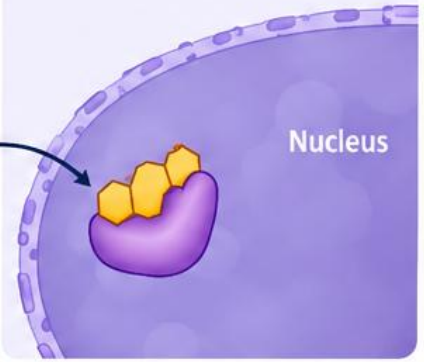
2 Bind to intracellular receptor

In the cytoplasm (or nucleus), the steroid binds to a specific receptor (e.g., androgen receptor, glucocorticoid receptor).



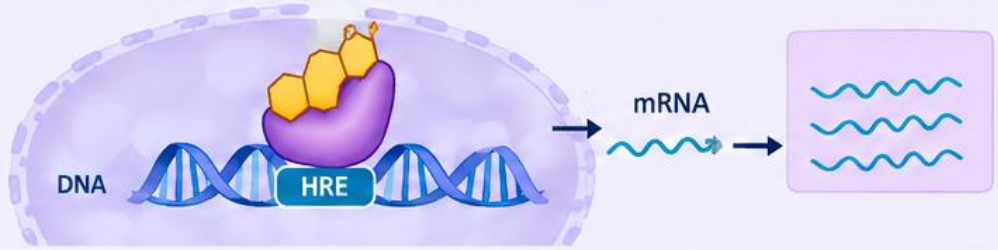
3 Translocate to nucleus

The steroid–receptor complex moves into the nucleus.



4 Bind to DNA and regulate gene expression

The complex binds to hormone response elements (HREs) on DNA, activating or repressing target genes.



Changes in gene expression

- Increased protein synthesis
- Enhanced cell growth and proliferation
- Altered metabolism
- Suppressed inflammation (for glucocorticoids)

5 Biological effects

The changes in gene expression lead to the steroid's effects, which depend on the type of steroid and target tissue.



Anabolic-androgenic steroids

- Increase muscle protein synthesis
- Promote muscle growth
- Support bone density



Glucocorticoids

- Reduce inflammation
- Suppress immune response
- Increase gluconeogenesis (raise blood glucose)



Other synthetic steroids

- e.g., progestins (affect reproductive system)
- e.g., mineralocorticoids (affect salt/water balance)



In summary: Synthetic steroids work by entering cells, *binding* to *intracellular* steroid receptors, modulating gene expression, and ultimately altering protein production and cellular function.

Adverse Drug Reactions (ADRs) of Synthetic Steroids

Synthetic steroids can cause a range of adverse effects, especially with high doses or long-term use, due to their systemic actions on multiple organs and tissues.

Endocrine & Reproductive



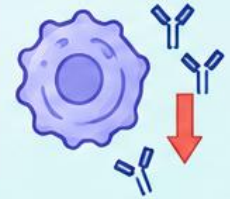
- **Suppression of HPA axis** (↓ endogenous cortisol)
- **Hypogonadism** (↓ testosterone, ↓ sperm count)
- **Gynecomastia** (males)
- **Menstrual irregularities** (females)
- **Virilization** (females) e.g., hirsutism, voice deepening

Metabolic



- Hyperglycemia / insulin resistance
- Dyslipidemia (↑ LDL, ↓ HDL)
- Weight gain / fat redistribution
- Fluid and sodium retention (glucocorticoids)

Immune System

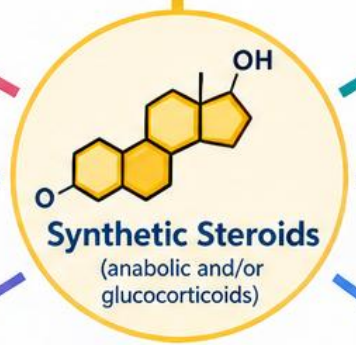


- **Immunosuppression** (↑ risk of infection)
- Delayed wound healing
- Masking of infection signs (e.g., fever, inflammation)

Gastrointestinal / Hepatic



- Hepatotoxicity (especially 17-α-alkylated steroids)
- Cholestasis
- Nausea, vomiting, abdominal pain



Psychiatric / Neurologic



- Mood changes (euphoria, irritability, aggression)
- Anxiety, depression
- Sleep disturbance / insomnia
- Headache, dizziness
- Rare: psychosis

Cardiovascular



- Hypertension
- Edema
- Thromboembolic events (especially with high doses/oral use)
- Possible cardiomyopathy (rare)

Skin & Others



- Acne
- Hirsutism
- Skin thinning, striae
- Easy bruising
- Delayed growth (children)

Musculoskeletal



- Osteoporosis (long-term use)
- Muscle atrophy and weakness (due to catabolism)
- Tendon rupture
- Aseptic necrosis of femoral head (high-dose, long-term)

Key Point:

The **risk and severity of ADRs** depend on the type, dose, duration of therapy, route of **administration,** and individual patient factors.

Use synthetic steroids only when indicated and monitor for adverse effects regularly.

1. Glucocorticoids

Main Actions

- Potent anti-inflammatory effects
- Immunosuppression
- Anti-allergic effects
- Regulation of carbohydrate, protein, and lipid metabolism

Common Clinical Uses

- Asthma
- COPD exacerbation
- Rheumatoid arthritis
- Systemic lupus erythematosus (SLE)
- Organ transplantation
- Cerebral edema
- Allergic reactions
- Dermatologic disorders (eczema, psoriasis)

Types or groups:

- **Short-acting**

  Hydrocortisone

  Cortisone
- **Intermediate-acting**

  Prednisone

  Prednisolone

  Methylprednisolone

  Triamcinolone
- **Long-acting**

  Dexamethasone

  Betamethasone

Drug	Relative Potency	Clinical Uses
Hydrocortisone	Short-acting	Adrenal insufficiency, replacement therapy
Prednisone	Intermediate	Autoimmune diseases, asthma, inflammatory disorders
Prednisolone	Intermediate	Rheumatoid arthritis, lupus, inflammatory bowel disease
Methylprednisolone	Intermediate	Severe inflammation, transplant rejection
Dexamethasone	Long-acting	Cerebral edema, COVID-19 inflammation, antiemetic in chemotherapy
Betamethasone	Long-acting	Fetal lung maturation, severe allergic disorders
Triamcinolone	Intermediate	Intra-articular injections, dermatologic diseases

2. Mineralocorticoids

Main Actions

- Increase sodium retention
- Increase water retention
- Increase potassium excretion
- Maintain blood pressure and extracellular fluid volume

Drugs

- Fludrocortisone
- Desoxycorticosterone acetate (DOCA)

Common Clinical Uses

- Addison's disease
- Congenital adrenal hyperplasia with salt wasting
- Orthostatic hypotension
- Adrenal crisis prevention

3. Androgenic Steroids

Main Actions: Activation of androgen receptor

- Promote protein synthesis
- Increase muscle mass
- Stimulate erythropoiesis
- Develop male secondary sexual characteristics

*****Misuse**

- Bodybuilding
- Athletic performance enhancement
- Muscle growth

Common Clinical Uses

- Male hypogonadism
- Delayed puberty
- Cachexia
- HIV-associated wasting
- Severe burns
- Certain anemias

Drugs	Clinical Uses
Testosterone enanthate	Male hypogonadism
Testosterone cypionate	Testosterone replacement therapy
Testosterone undecanoate	Long-term androgen replacement
Nandrolone decanoate	Anemia, muscle wasting disorders
Oxandrolone	Burns, trauma, weight loss syndromes
Stanozolol	Hereditary angioedema (limited use)
Methandrostenolone	Historical anabolic use
Oxymetholone	Aplastic anemia

4. Synthetic Estrogens

Main Actions

- Activate estrogen receptors
- Development and maintenance of female reproductive tissues
- Regulation of menstrual cycle
- Prevention of bone loss

Common Clinical Uses

- Menopausal hormone replacement therapy (HRT)
- Combined oral contraception
- Hypogonadism in females
- Gender-affirming hormone therapy
- Osteoporosis prevention

Drug	Clinical Uses
Ethinyl estradiol	Oral contraceptives
Estradiol valerate	Hormone replacement therapy
Estradiol cypionate	Menopausal symptoms
Conjugated estrogens	Menopausal hormone therapy
Diethylstilbestrol (DES)	Historical use, rarely used today

5. Synthetic Progestins

Main Actions

- Mimic progesterone
- Suppress ovulation
- Reduce endometrial proliferation
- Maintain pregnancy

Common Clinical Uses

- Oral contraceptives
- Emergency contraception
- Endometriosis
- Abnormal uterine bleeding
- Endometrial hyperplasia
- Assisted reproductive technology
- Appetite stimulation in cancer patients

Drug	Generation	Clinical Uses
Medroxyprogesterone acetate	First	Contraception, endometrial disorders
Megestrol acetate	First	Cancer cachexia, appetite stimulation
Norethindrone	First	Oral contraceptives
Levonorgestrel	Second	Emergency contraception, IUDs
Desogestrel	Third	Oral contraception
Drospirenone	Fourth	Contraception, PMDD
Dienogest	Fourth	Endometriosis

6. Progestins (Synthetic Progestogens)

Main action: Bind progesterone receptors, suppress ovulation, alter cervical mucus, and regulate endometrial growth

Major Drugs

- Medroxyprogesterone acetate
- Megestrol acetate
- Norethindrone
- Levonorgestrel
- Desogestrel
- Drospirenone
- Dienogest
- Etonogestrel

6. Progestins (Synthetic Progestogens)

Clinical Uses:

Contraception

- Combined oral contraceptives
- Progestin-only pills
- Implants
- Hormonal IUDs

Endometriosis

- Dienogest
- Medroxyprogesterone acetate

Abnormal Uterine Bleeding

- Dysfunctional uterine bleeding
- Endometrial hyperplasia prevention

Infertility Treatment

- Luteal phase support

Cancer/Palliative Care

- Megestrol acetate for cancer cachexia and appetite stimulation

Summary of Synthetic Steroid Classes

Class	Representative Drugs	Main Clinical Uses
Glucocorticoids	Prednisone, Dexamethasone, Methylprednisolone	Anti-inflammatory, immunosuppression, asthma, autoimmune diseases
Mineralocorticoids	Fludrocortisone	Adrenal insufficiency, orthostatic hypotension
Androgens	Testosterone, Methyltestosterone	Hypogonadism, delayed puberty, gender-affirming therapy
Anabolic Steroids	Nandrolone, Oxandrolone, Oxymetholone	Muscle wasting, burns, anemia
Estrogens	Ethinyl estradiol, Estradiol	Menopause, contraception, osteoporosis prevention
Progestins	Medroxyprogesterone, Levonorgestrel, Dienogest	Contraception, endometriosis, abnormal uterine bleeding

PHARMACOLOGY OF SYNTHETIC STEROIDS

From Molecules to Mechanisms. From Knowledge to Better Care.

- WIDELY USED IN CLINICAL PRACTICE
- POWERFUL ANTI-INFLAMMATORY AND IMMUNOSUPPRESSIVE AGENTS
- CRUCIAL IN MANY ACUTE AND CHRONIC CONDITIONS
- USE JUDICIOUSLY. MONITOR CAREFULLY. PRESCRIBE WISELY.

KEY DRUG MECHANISMS

1. **GLUCOCORTICOID RECEPTOR ACTIVATION**
   Binds to intracellular glucocorticoid receptors in target cells.
2. **GENE TRANSCRIPTION MODULATION**
   Regulates transcription of target genes by binding to GREs (glucocorticoid response elements).
3. **ANTI-INFLAMMATORY EFFECTS**
   Inhibits phospholipase A_2, ↓ prostaglandins, leukotrienes and other inflammatory mediators.
4. **IMMUNOSUPPRESSION**
   Decreases T-lymphocyte proliferation, cytokine production and immune cell activity.
5. **CYTOKINE INHIBITION**
   Suppresses pro-inflammatory cytokines (IL-1, IL-2, IL-6, TNF-α, IFN-γ).
6. **METABOLIC REGULATION**
   Alters glucose, protein and lipid metabolism; maintains vascular reactivity.

MAJOR ADVERSE EFFECTS

1. **HYPERGLYCEMIA / DIABETES**
   Increases gluconeogenesis and insulin resistance; may precipitate or worsen diabetes.
2. **OSTEOPOROSIS**
   Decreases bone formation and increases bone resorption → ↑ risk of fractures.
3. **ADRENAL SUPPRESSION**
   Suppresses HPA axis → risk of adrenal insufficiency on abrupt withdrawal.
4. **CUSHINGOID FEATURES**
   Weight gain, moon face, central obesity, striae, acne, hirsutism.
5. **INFECTION RISK**
   Impaired immune response → increased susceptibility to infections.
6. **HYPERTENSION**
   Sodium and water retention; ↑ blood pressure and cardiovascular risk.





CLINICAL PEARLS

- Use the lowest effective dose for the shortest duration.
- Monitor for adverse effects and adjust therapy.
- Taper steroids to avoid adrenal crisis.

POWERFUL BENEFITS WHEN USED WISELY. | BALANCE EFFICACY WITH SAFETY.

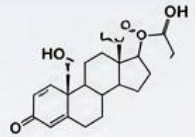
PHARMACOLOGY OF SYNTHETIC STEROIDS

Structure–Mechanism–Types–Clinical Applications

1. GLUCOCORTICOIDS

Examples

Hydrocortisone, Prednisone, Prednisolone, Methylprednisolone, Dexamethasone, Betamethasone, Triamcinolone, Budesonide, Beclomethasone



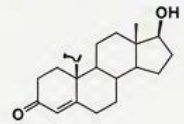
Main Actions

Anti-inflammatory, Immunosuppressive, Anti-allergic, Anti-proliferative

2. MINERALOCORTICOIDS

Examples

Fludrocortisone, Desoxycorticosterone acetate (DOCA)



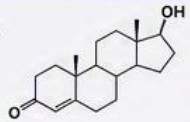
Main Actions

Na^+ and water retention, K^+ excretion, ↑ Blood volume, ↑ Blood pressure

3. ANDROGENS

Examples

Testosterone, Testosterone enanthate, Testosterone cypionate, Testosterone undecanoate, Methyltestosterone, Dihydrotestosterone (DHT)



Main Actions

Development & maintenance of male reproductive organs & secondary sexual characteristics; ↑ Libido

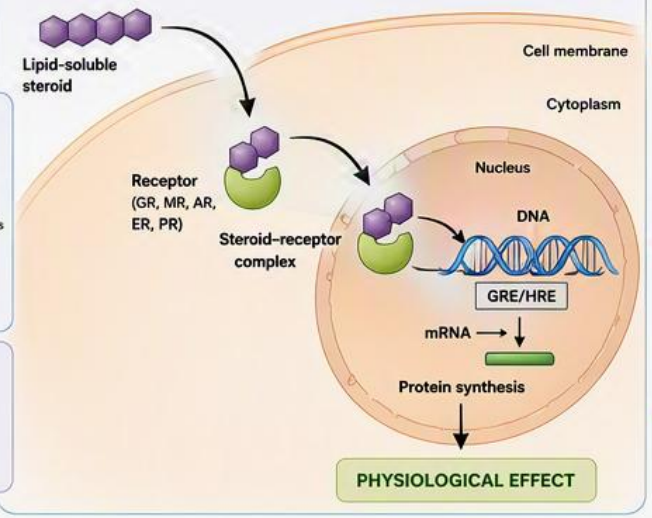
MECHANISM HIGHLIGHTS

- Diffuses across cell membrane
- Binds to intracellular receptor
- Complex translocates to nucleus
- Binds to hormone response elements (GRE/HRE) on DNA
- Regulates gene transcription
- Alters protein synthesis → Physiological effects

KEY POINTS

- Onset: Hours
- Duration: Long (hrs–days)
- Most actions are genomic
- Wide tissue distribution

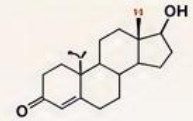
CENTRAL MECHANISM OF ACTION (Genomic Pathway)



4. ANABOLIC–ANDROGENIC STEROIDS (AAS)

Examples

Nandrolone, Oxandrolone, Oxymetholone, Danazol, Stanozolol, Methandienone, Fluoxymesterone, Boldenone



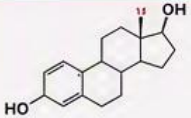
Main Actions

Anabolic (↑ protein synthesis), ↑ lean body mass, ↑ bone formation, ↑ erythropoiesis; Moderate androgenic effects

5. ESTROGENS

Examples

Ethinyl estradiol, Estradiol valerate, Estradiol cypionate, Conjugated estrogens, Estradiol (transdermal)



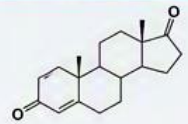
Main Actions

Development & maintenance of female reproductive organs; secondary sexual characteristics; bone protection; lipid profile modulation

6. PROGESTINS (SYNTHETIC PROGESTOGENS)

Examples

Medroxyprogesterone acetate, Levonorgestrel, Norethindrone, Dienogest, Drospirenone, Etonogestrel, Megestrol acetate



Main Actions

Prepares endometrium, thickens cervical mucus, suppresses ovulation, anti-proliferative in endometrium

CLINICAL APPLICATIONS (Illustrative)

1. GLUCOCORTICOIDS


Asthma / COPD


Rheumatoid Arthritis


SLE (Lupus)


Inflammatory Bowel Disease


Cerebral Edema (Brain tumors)


Severe Allergic Reactions


Leukemia / Lymphoma


Autoimmune Disorders


Adrenal Insufficiency (Addison's disease)

Key Uses: Anti-inflammatory, Immunosuppression, Anti-allergic, Replacement therapy

2. MINERALOCORTICOIDS



Adrenal Insufficiency (Addison's disease)

- Salt wasting
- Hyponatremia
- Hyperkalemia
- Hypotension



Orthostatic / Postural Hypotension



Key Uses: Na^+ & water retention in adrenal disorders, Increase blood volume & blood pressure

3. ANDROGENS



Hypogonadism in Males

- ↓ Libido
- Erectile dysfunction
- Infertility
- ↓ Muscle mass
- Fatigue



Delayed Puberty (Constitutional)

Gender-Affirming Therapy (Masculinizing)

Key Uses: Replacement therapy, Pubertal induction, Gender-affirming therapy

4. ANABOLIC–ANDROGENIC STEROIDS (AAS)



Muscle Wasting / Cachexia (HIV, Cancer)

Severe Burns Recovery



Aplastic Anemia / Chronic Anemia



Osteoporosis (↑ Bone Mass)





Hereditary Angioedema (Danazol)

Key Uses: ↑ Lean body mass, Promote anabolism, Treat wasting, anemia, burns, osteoporosis

5. ESTROGENS



Menopausal Hormone Replacement Therapy

- Hot flashes
- Night sweats
- Mood changes
- Urogenital atrophy



Osteoporosis Prevention

Hypogonadism / Premature Ovarian Insufficiency



Gender-Affirming Therapy (Feminizing)



Key Uses: Menopause, Bone protection, Hormonal deficiency, Feminizing therapy

6. PROGESTINS (SYNTHETIC PROGESTOGENS)

Contraception & Family Planning



Endometriosis



Abnormal Uterine Bleeding



Endometrial Hyperplasia Prevention





Cancer Cachexia / Appetite Stimulation (Megestrol acetate)

Key Uses: Contraception, Endometriosis, AUB, Luteal support, Cachexia

CLASS EFFECTS / ADVERSE EFFECTS (Common)

- Metabolic: Hyperglycemia, dyslipidemia
- Fluid/Electrolyte: Na^+ retention, K^+ loss, edema, HTN (mineralocorticoids, glucocorticoids)
- Musculoskeletal: Osteoporosis, myopathy
- GI: Ulcers, dyspepsia • Skin: Acne, striae, thinning
- Endocrine: HPA axis suppression, Cushingoid features, growth suppression in children
- Neuropsychiatric: Mood changes, insomnia, psychosis

IMPORTANT PHARMACOKINETIC CONSIDERATIONS


Lipid soluble → wide tissue distribution


High protein binding (especially AAS)

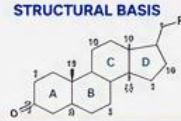

Metabolized in liver (CYP3A4)


Excreted in urine (as metabolites)

GENERAL PEARLS

- ☑ Use the lowest effective dose for the shortest duration.
- ☑ Do not stop glucocorticoids abruptly → risk of adrenal crisis.
- ☑ Monitor: BP, blood glucose, weight, bone density, eye pressure.
- ☑ Beware of drug interactions (CYP3A4 inducers/inhibitors).

STRUCTURAL BASIS



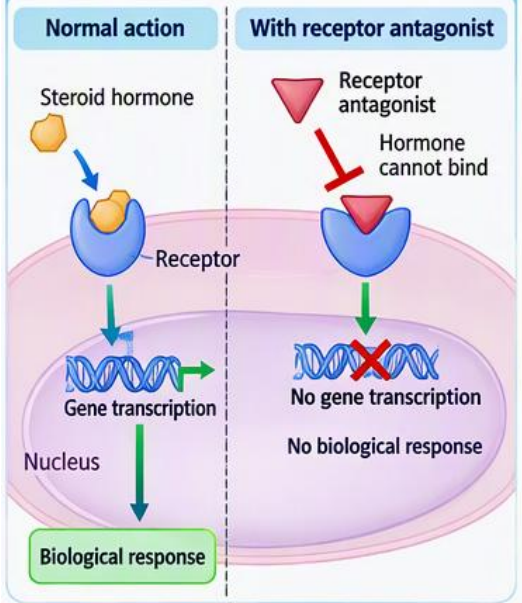
All steroids share the cyclopentanoperhydrophenanthrene nucleus derived from cholesterol.

Steroid Receptor Antagonists

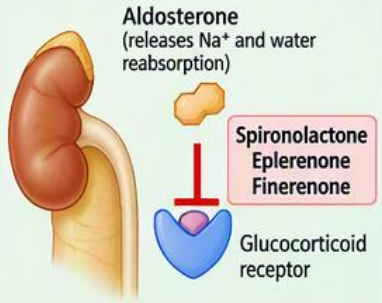
Block the action of endogenous steroid hormones by **competitively inhibiting their receptors.**

- **Hormone levels:** usually unchanged or increased
- **Onset of action:** relatively rapid
- **Main use:** endocrine disorders, hormone-dependent cancers, reproductive conditions

Mechanism of action (general)



Mineralocorticoid Receptor Antagonists



Drugs

- Spironolactone
- Eplerenone
- Finerenone

Pharmacological effects

- Block aldosterone receptors in distal nephron
- ↓ Na^+ and water reabsorption
- ↑ K^+ retention (potassium-sparing diuretic)
- Reduce cardiac and renal fibrosis

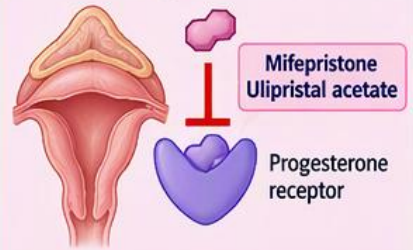
Clinical uses

- Heart failure (HFrEF)
- Resistant hypertension
- Primary hyperaldosteronism (Conn syndrome)
- Edema due to liver cirrhosis (ascites)
- Chronic kidney disease (finerenone)
- Hirsutism and PCOS (spironolactone)

Adverse effects

- Hyperkalemia
- Gynecomia
- Gynecomastia (esp. spironolactone)
- Impotence

Glucocorticoid Receptor Antagonist



Drugs

- Mifepristone
- Ulipristal acetate (partial antagonist)

Pharmacological effects

- Inhibit progesterone receptors
- Cause endometrial breakdown
- Increase uterine contractility
- Prevent implantation

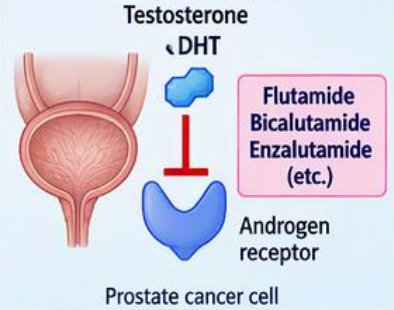
Clinical uses

- Medical abortion (with misoprostol)
- Emergency contraception (ulipristal)
- Uterine fibroids (selected indications)

Adverse effects

- Vaginal bleeding
- Abdominal pain
- Nausea
- Headache

Androgen Receptor Antagonists



Drugs

- Flutamide
- Bicalutamide
- Nilutamide
- Enzalutamide
- Apalutamide
- Darolutamide

Pharmacological effects

- Block androgen receptors
- Inhibit actions of testosterone and DHT

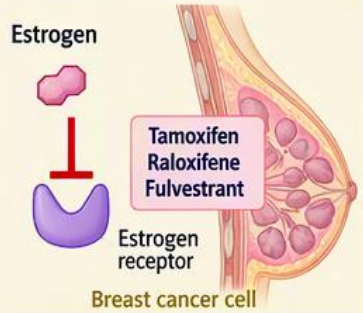
Clinical uses

- Prostate cancer
- Metastatic castration-resistant prostate cancer
- Precocious puberty (selected cases)

Adverse effects

- Gynecomastia
- Hot flashes
- Reduced libido
- Hepatotoxicity (especially flutamide)

Estrogen Receptor Antagonists



Drugs

Selective estrogen receptor modulators (SERMs): Tamoxifen, Raloxifene

- Pure antagonist: Fulvestrant

Pharmacological effects

- Block estregen receptors
- Tamoxifen and raloxifene: antagonist in some tissues and agonist in others (SERMs)
- Fulvestrant: receptor blockade and degradation

Clinical uses

- Estrogen receptor-positive breast cancer (treatment and prevention)
- Osteoporosis prevention (raloxifene)
- Metastatic hormone receptor-positive breast cancer (fulvestrant)

Adverse effects

- Hot flashes
- Venous thromboembolism
- Endometrial cancer risk (tamoxifen)

Key point: Steroid receptor antagonists do not reduce hormone synthesis; they block the hormone's action at the receptor.
They are widely used in endocrine disorders, hormone-dependent cancers, and reproductive conditions.

STEROID SYNTHESIS INHIBITORS

Reduce the production of steroid hormones by inhibiting enzymes involved in steroidogenesis

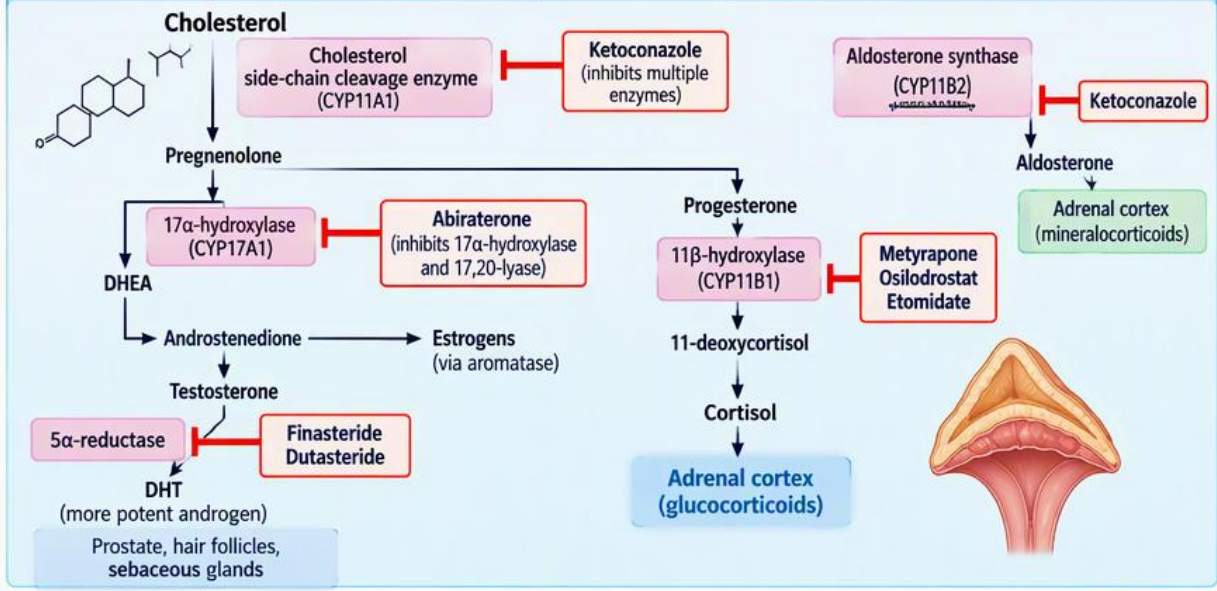
- **Hormone levels:** decreased
- **Onset of action:** variable (usually slower than receptor antagonists)
- **Main use:** endocrine disorders and hormone-dependent cancers

Mechanism of action (general)

- Inhibit key enzymes in steroid biosynthesis (mainly in adrenal glands, gonads, or peripheral tissues)
- Decrease synthesis of glucocorticoids, mineralocorticoids, and/or sex steroids
- Lower circulating hormone levels
- Used in hormone-dependent disorders and cancers

Unlike receptor antagonists, these drugs reduce hormone production.

Overview of steroid synthesis pathway and sites of inhibition



Sites of steroid synthesis



Adrenal glands
- Cortisol
- Aldosterone
- Adrenal androgens (DHEA, androstenedione)



Testes
- Testosterone
- DHT
- Estradiol (small amount)

Testes



Ovaries
- Estrogens
- Progesterone
- Androgens (small amount)



Peripheral tissues
(skin, adipose tissue, prostate tumor)
- Conversion (e.g., 5α-reductase, aromatase)

Major classes of steroid synthesis inhibitors

	Ketoconazole (broad inhibitor)	Metyrapone	Osilodrostat	Mitotane	Etomidate (IV)	Abiraterone	5α-Reductase inhibitors	
							Finasteride (type II)	Dutasteride (type I and II)
Target / Mechanism	Inhibits multiple CYP450 enzymes: cholesterol side-chain cleavage (CYP11A1), 17α-hydroxylase, 11β-hydroxylase	Inhibits 11β-hydroxylase (CYP11B1)	Potent inhibitor of 11β-hydroxylase	Adrenolytic agent (destroys adrenal cortical cells)	Inhibits 11β-hydroxylase (CYP11B1)	Irreversibly inhibits CYP17A1 (17α-hydroxylase and 17,20-lyase)	Inhibits 5α-reductase (conversion of testosterone to DHT)	
Hormones decreased	Cortisol, androgens, aldosterone	Cortisol	Cortisol	Cortisol (and other adrenal steroids)	Cortisol	Androgens (testes, adrenal, prostate tumor)	Dihydrotestosterone (DHT)	
Clinical uses	Cushing syndrome (when surgery not possible) Androgen-dependent disorders (e.g., prostate cancer – limited use)	Cushing syndrome Diagnostic test for adrenal insufficiency	Cushing disease (adults)	Adrenocortical carcinoma Severe Cushing syndrome	Severe hypercortisolism (e.g., Cushing crisis) when oral therapy not possible	Metastatic prostate cancer Castration-resistant prostate cancer	Benign prostatic hyperplasia (BPH) Male pattern baldness	
Adverse effects	Hepatotoxicity, gynecomastia, GI disturbance, drug interactions	Hypertension, edema, hirsutism (↑ ACTH and androgens), GI upset	Adrenal insufficiency, hypokalemia, QT prolongation	GI toxicity, neurologic symptoms (ataxia, confusion), adrenal insufficiency	Adrenal insufficiency, hypotension (with prolonged) use), nausea	Hypertension, fluid retention, hypokalemia (↑ mineralocorticoids), hepatotoxicity (requires steroids)	Erectile dysfunction, decreased libido, ejaculatory disorders	

Clinical take-home messages

- Steroid synthesis inhibitors lower hormone levels by blocking steroidogenic enzymes, whereas receptor antagonists block hormone action at the receptor.
- They are important in Cushing syndrome, adrenocortical carcinoma, prostate cancer, and BPH.
- Adverse effects often reflect loss of steroid hormones and/or accumulation of upstream precursors (e.g., hypertension, hypokalemia, hirsutism).

Mnemonic (examples)

- **"MET"** → Metyrapone, Etomidate, Mitotane (11β-hydroxylase / adrenal)
- **"Keto"** → Ketoconazole (multiple steps)
- **"ABIR"** → Abiraterone (CYP17A1)
- **"FIN-DUT"** → Finasteride, Dutasteride (5α-reductase)